Paddy Yield Prediction

IE 498: Machine Learning in Finance

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1 Introduction

Rice is a major food source worldwide, and understanding the factors associated with its production can support agricultural planning. Machine learning provides a way to examine relationships between agricultural conditions, farming practices, and crop production. This project uses the Paddy Dataset from the UCI Machine Learning Repository [Subramaniyan, 2023] to predict paddy yield from agricultural and environmental features. We compare different machine learning models and investigate whether dimensionality reduction improves predictive performance.

Four models are considered: Linear Regression, Lasso Regression, Random Forest Regression, and Support Vector Regression (SVR). Principal Component Analysis (PCA) is also applied to address multicollinearity among the predictors. The analysis proceeds from exploratory data analysis (EDA) to preprocessing, model fitting, hyperparameter tuning, and performance comparison.

All models and preprocessing were implemented using scikit-learn [Pedregosa et al., 2011]. The modeling and evaluation workflow follows standard machine learning practices discussed in [Raschka, Sebastian and Mirjalili, Vahid, 2019].

2 Data Understanding

The paddy yield dataset contains 2,789 observations and 44 input features, with `Paddy yield(in Kg)` as the target. The task is supervised regression because the objective is to predict a numerical quantity. The dataset contains no missing values.

The input variables describe agricultural practices and environmental conditions. Agricultural variables include cultivated area, nursery area, seed rate, fertilizer applications, and pest-control measures. Environmental variables include rainfall, irrigation, temperature, wind, and humidity. Categorical variables include agricultural block, paddy variety, soil type, and wind direction.

3 Exploratory Data Analysis

3.1 Target Variable

Figure 1 reveals a wave-like distribution for paddy yield rather than a smooth curve. This may stem from underlying subpopulations, categorical separations, diverse agricultural conditions, or data-collection artifacts. To investigate the cause, we next examine the relationship between the features and the target variable.

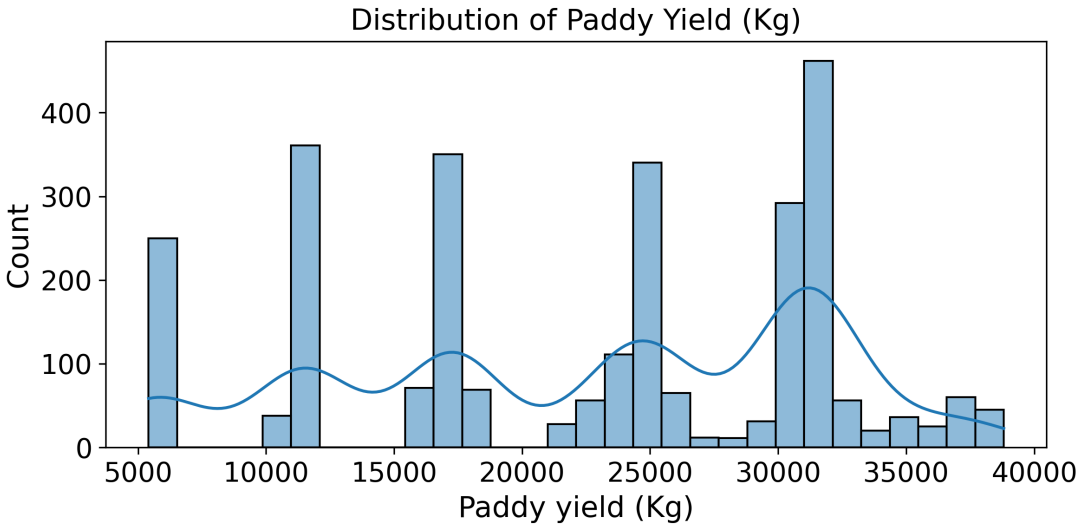


Figure 1: Distribution of paddy yield in kilograms.

3.2 Numerical Features

The distributions of continuous and integer-valued predictors are shown in Figures 2 and 3, respectively. Some predictors have distributions similar to that of the target variable or to each other. Therefore, we further investigate their relationships using correlation.

Figures 4 and 5 present the feature-to-feature correlation heatmap and Pearson correlations with paddy yield, respectively. The feature correlation matrix reveals strong collinearity among several features, indicating overlapping information in the feature space. Meanwhile, the target correlations show that yield is positively correlated with some agricultural inputs, such as fertilizer and pest control applications (`Urea_40Days`, `Potassh_50Days`, `DAP_20days`, `Micronutrients_70Days`, `Pest_60Day_(in_ml)`), cultivated area (`Hectares`), seed rate (`Seedrate_(in_Kg)`), nursery area (`LP_nurseryarea_(in_Tonnes)`), main field input (`LP_Mainfield_(in_Tonnes)`), and trash (`Trash_(in_bundles)`).

Moreover, Figure 6 shows scatterplots and fitted regression lines for features whose absolute

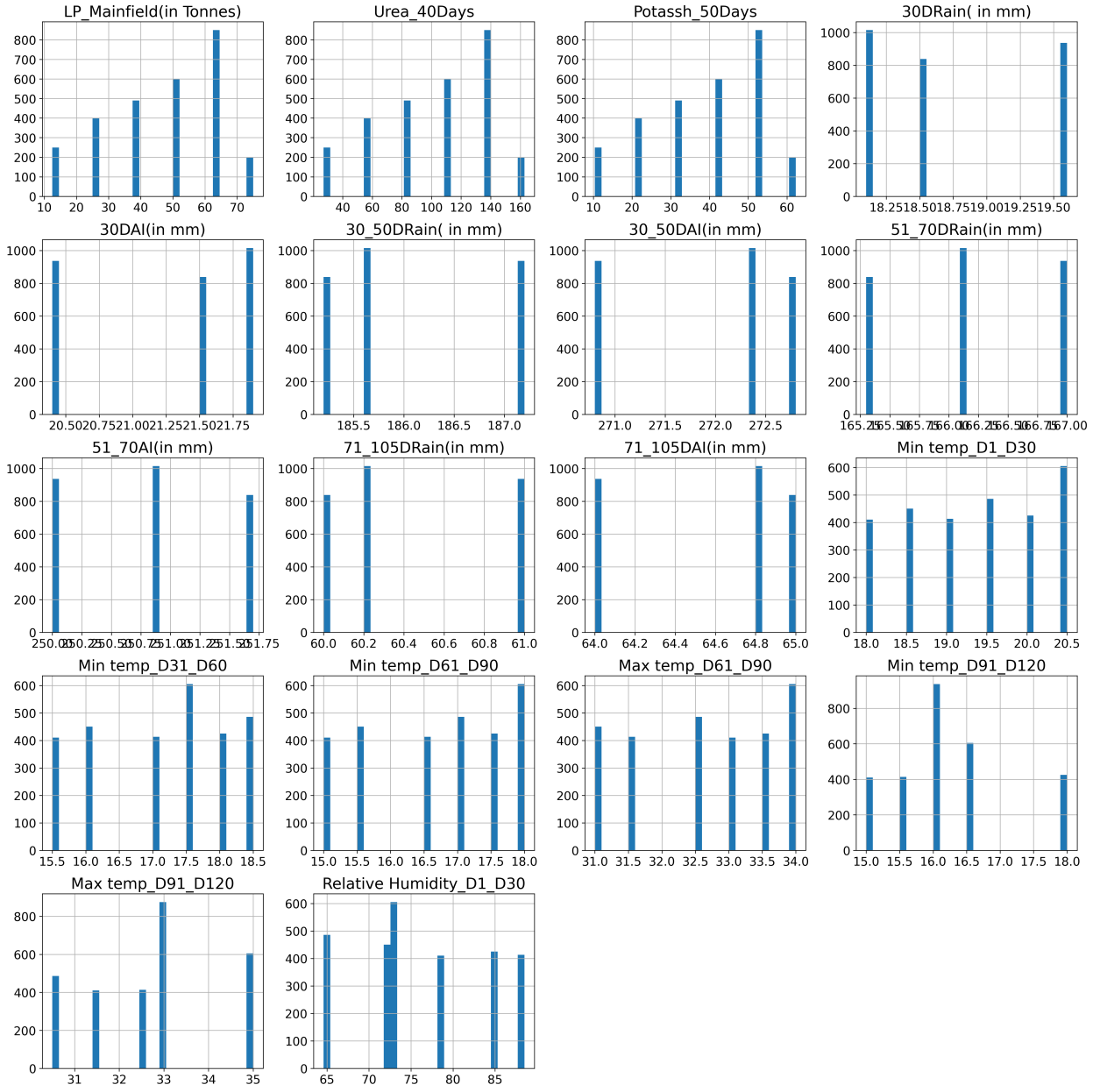


Figure 2: Distributions of continuous numerical features.

Pearson correlation with the target is higher than 0.8. The plots suggest approximately linear relationships for several important numerical predictors. This observation motivated the use of Linear Regression as a baseline model in Section 5, while the correlations among predictors motivated investigating Lasso and PCA.



Figure 3: Distributions of integer-valued features.

3.3 Categorical Features

Figure 7 summarizes the distributions of categorical features. The category counts are not uniform, so some categories are represented more frequently than others. Figure 8 compares paddy yield across the categories using boxplots. No clear pattern consistently separates the target based on these categorical features alone. Therefore, the EDA does not suggest that categorical variables alone explain the wave-like distribution of the target.

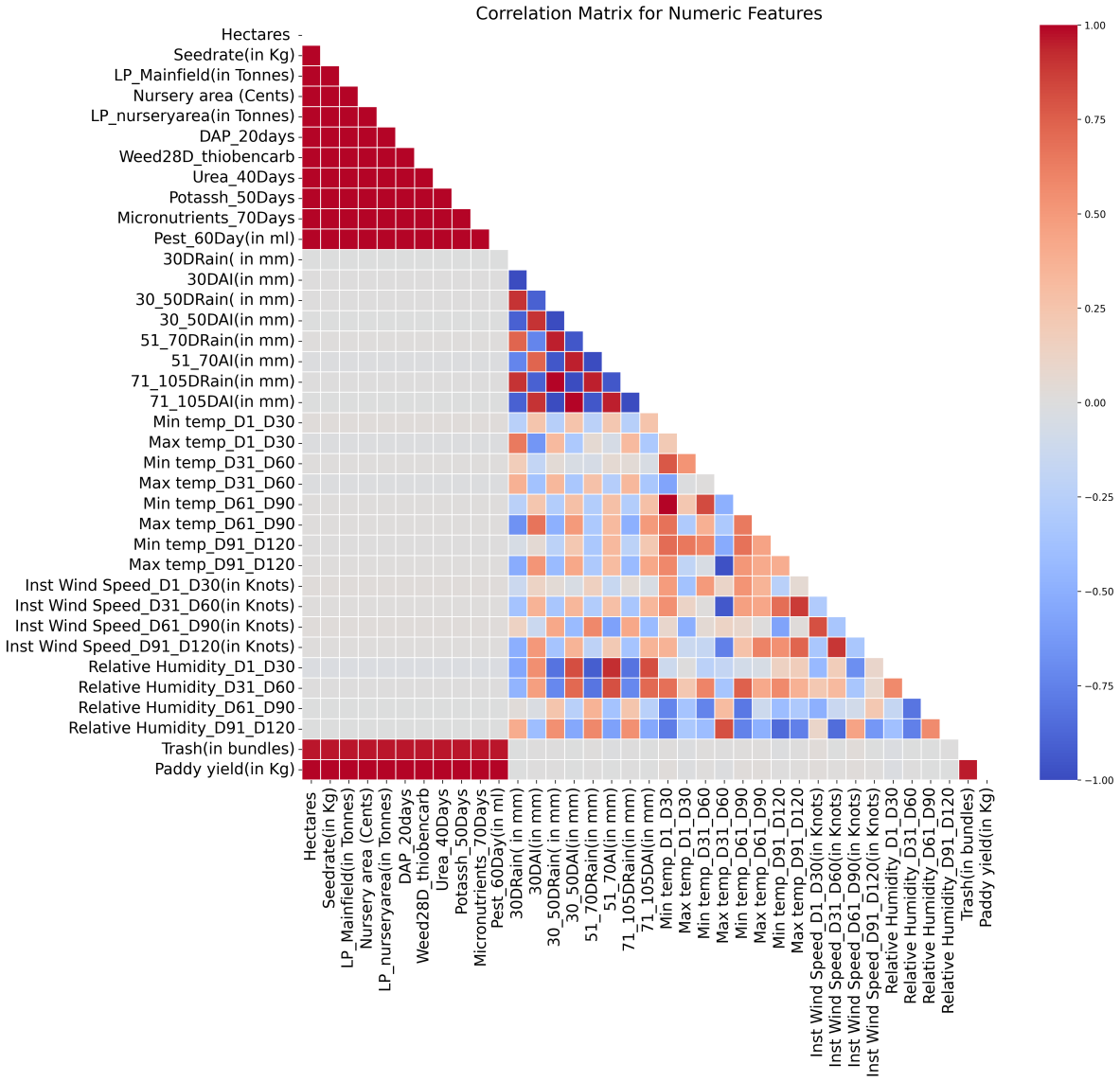


Figure 4: Correlation matrix of numerical features.

3.4 EDA Summary

The EDA identifies three main patterns. First, paddy yield has a wave-like distribution. Second, several numerical features have strong positive relationships with yield, and scatterplots suggest approximately linear relationships. Third, numerical predictors are strongly correlated with one another, indicating multicollinearity. These observations motivate comparing linear and nonlinear models and investigating whether regularization or dimensionality reduction improves performance.

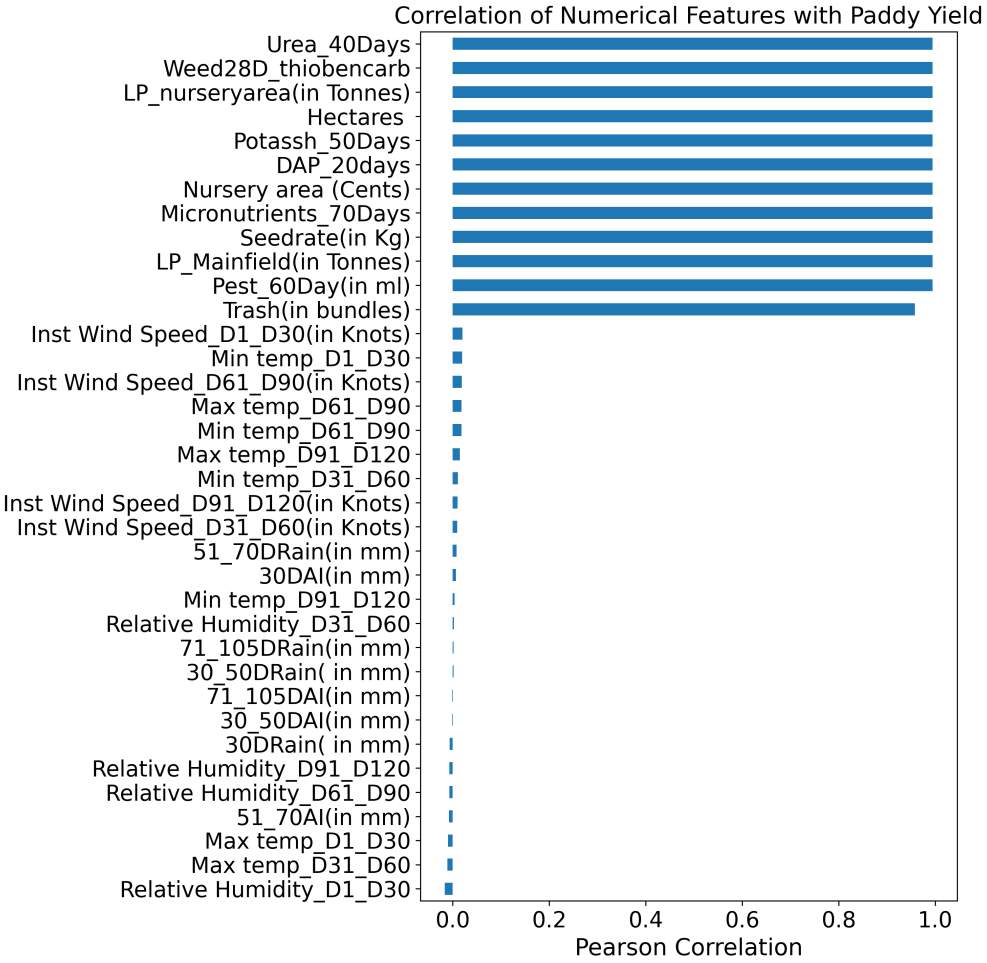


Figure 5: Pearson correlations between numerical features and paddy yield.

4 Preprocessing and Dimensionality Reduction

4.1 Data Preprocessing

The dataset was split into training and testing sets using an 80/20 split. Numerical predictors were standardized, and categorical predictors were converted into dummy variables. Transformations were fitted on the training set and then applied to the test set to avoid estimating preprocessing parameters from test data. After preprocessing, the feature space contained 71 columns. This processed feature matrix was used directly for the models without PCA in Section 5.

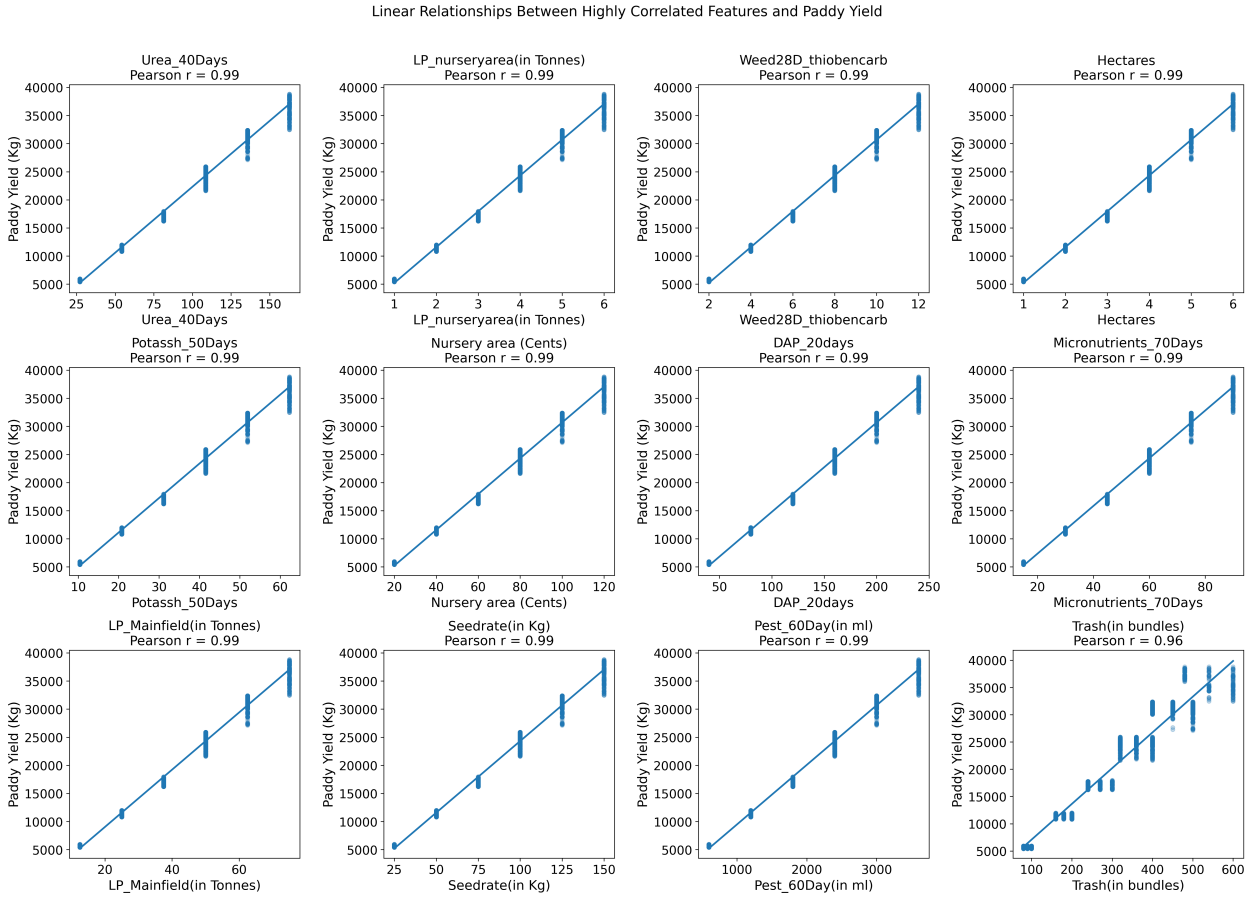


Figure 6: Relationships between highly correlated numerical features and paddy yield.

4.2 Principal Component Analysis (PCA)

Because the EDA revealed substantial multicollinearity, PCA was used to transform correlated predictors into orthogonal principal components. As shown in Figure 9, six components retained at least 95% of the variance in the processed training predictors, reducing the feature space from 71 columns to six. PCA was fitted on the training predictors only, and the same transformation was applied to the test predictors.

5 Paddy Yield Prediction

This section evaluates four models: Linear Regression, Lasso Regression, Random Forest, and Support Vector Regression (SVR). Each model is trained and evaluated both with and without PCA, resulting in eight model configurations. We evaluate performance using RMSE, MAE, R^2, and training time to assess predictive accuracy and computational efficiency.

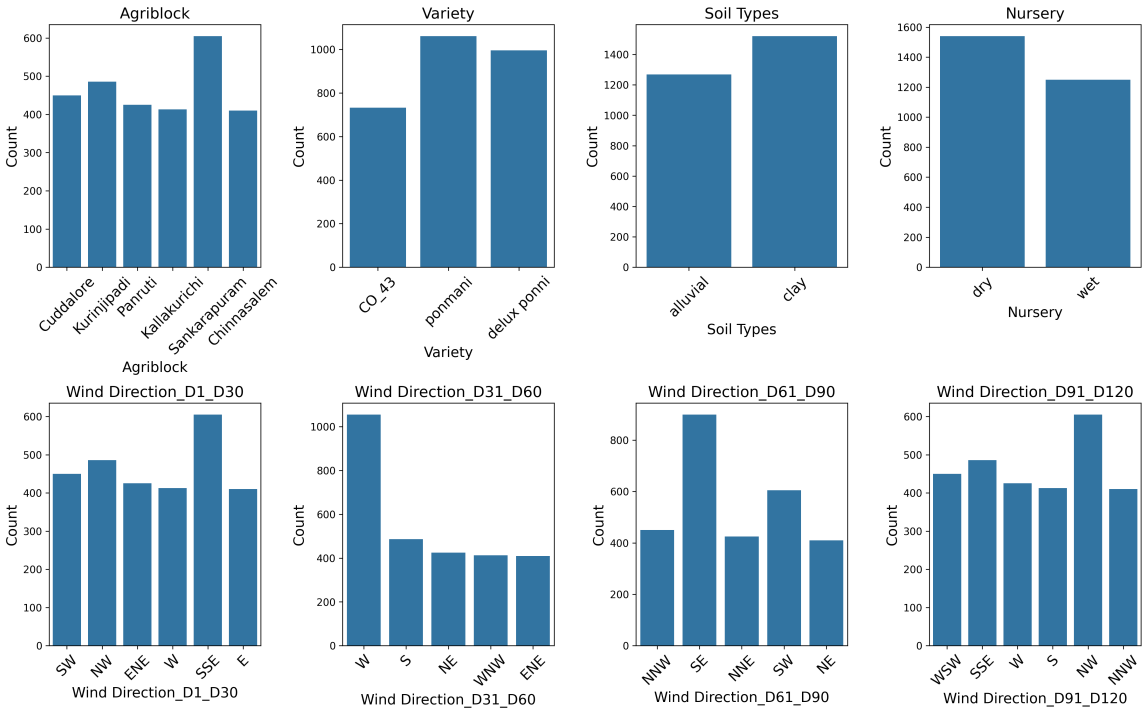


Figure 7: Distributions of categorical features.

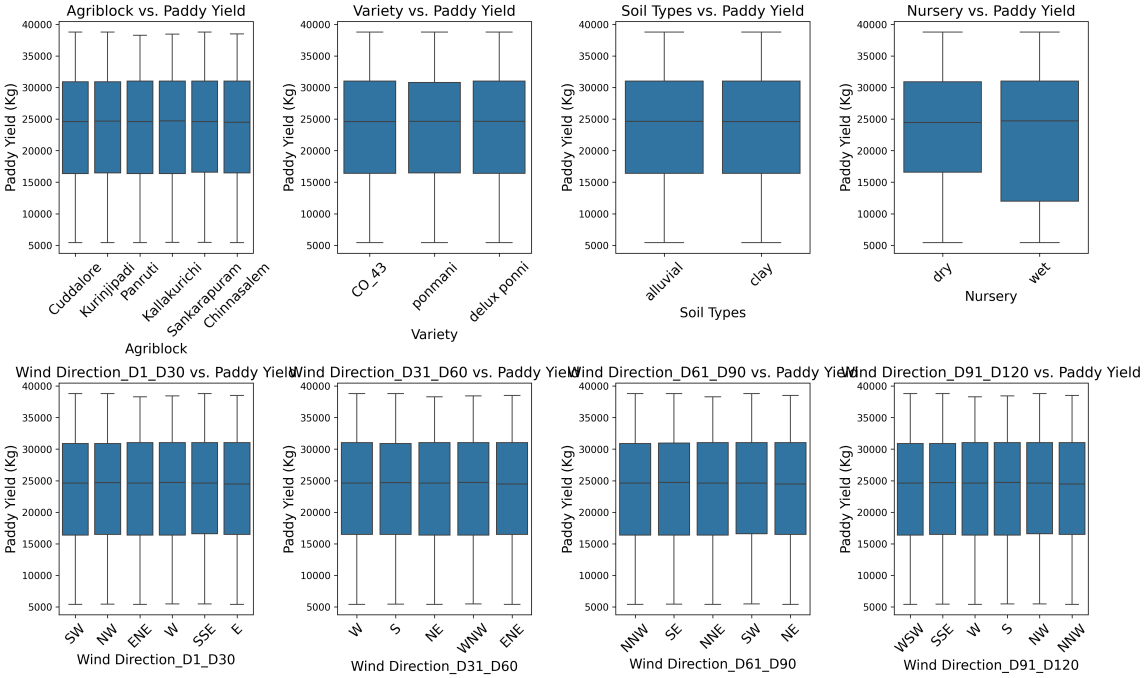


Figure 8: Paddy yield across categories of categorical features.

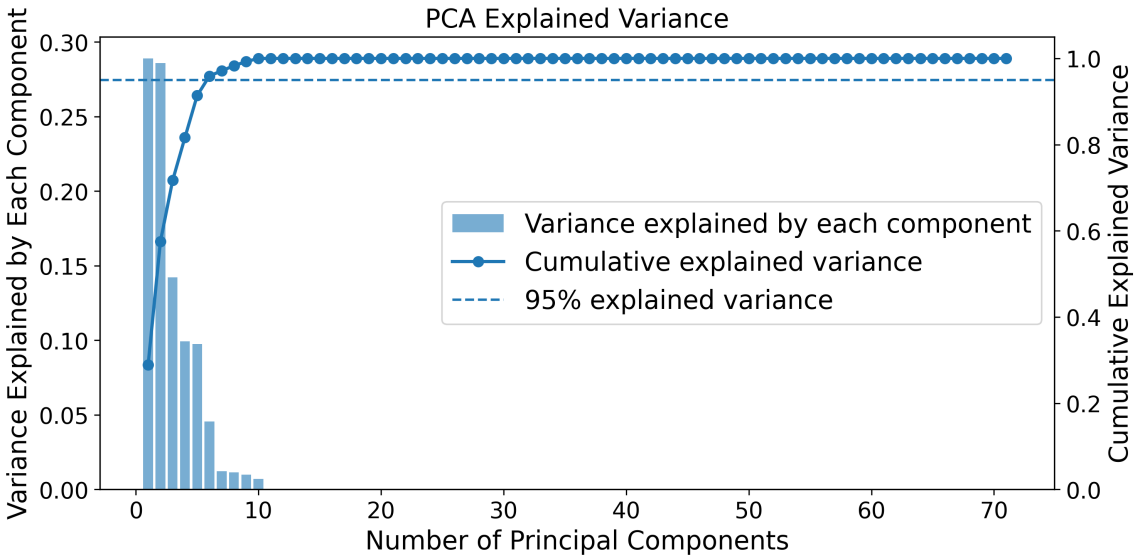


Figure 9: Explained variance and cumulative explained variance of PCA.

5.1 Linear Regression

Linear Regression was used as the baseline model because it is simple and easy to interpret. The model was fitted with and without PCA. Without PCA, it achieved an RMSE of 927.9360 kg, an MAE of 698.3192 kg, and an R^2 of 0.9894. With PCA, the RMSE increased to 1004.9139 kg and R^2 decreased to 0.9875.

Figure 10 shows the features with the largest absolute coefficients in the model without PCA. Features such as `Trash(in bundles)`, `Seedrate(in Kg)`, and `Hectares` have large coefficient magnitudes. These coefficients indicate conditional associations in the fitted model and should not be interpreted as causal effects.

5.2 Lasso Regression

Lasso adds an L_1 penalty to the least-squares objective, shrinking some coefficients to zero and thereby selecting features. We use five-fold cross-validation to select the penalty parameter α. Figures 11 and 12 show the cross-validation RMSE across values of α for the models without and with PCA, respectively. PCA changes the feature representation, so the optimal regularization parameter may differ between the two models.

Figure 13 shows the selected coefficients of the Lasso regression without PCA, with the penalty parameter $\alpha = 10.5729$. This model retained six nonzero coefficients. `Hectares` had the largest absolute coefficient. If the target represents total production, a larger cultivated area can naturally be associated with greater production.

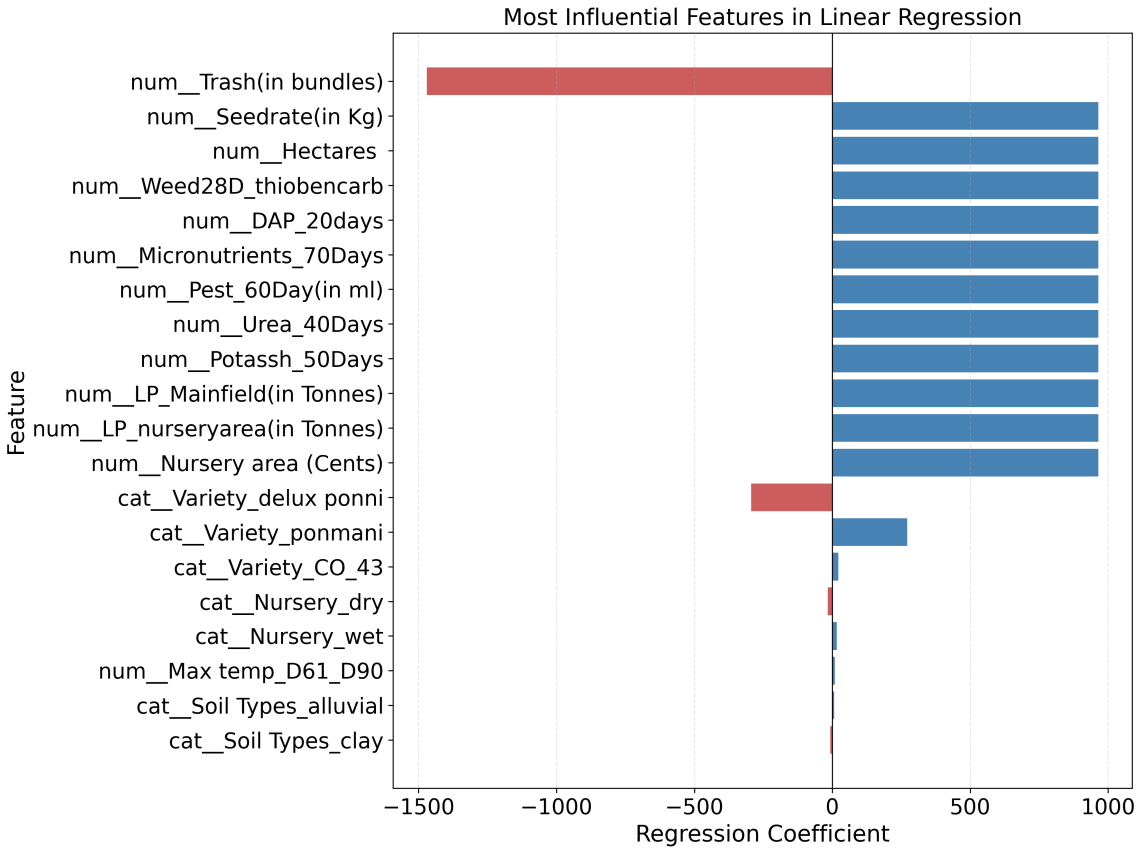


Figure 10: Most influential features in Linear Regression without PCA.

With $\alpha = 24.8591$, Lasso with PCA achieved an RMSE of 941.0506 kg, an MAE of 706.5258 kg, and an R^2 of 0.9891. It did not improve test-set accuracy over the Linear Regression baseline in this experiment.

5.3 Random Forest Regression

Random Forest combines multiple decision trees and can capture nonlinear relationships between predictors and the target. Without PCA, it achieved an RMSE of 886.8067 kg, an MAE of 632.5830 kg, and an R^2 of 0.9903. With PCA, its RMSE was 887.6489 kg and its R^2 remained 0.9903.

Figure 14 shows the feature importance ranking for Random Forest without PCA. We see that `LP_Mainfield(in Tonnes)`, `Seedrate(in Kg)`, and `Hectares` were among the most influential predictors.

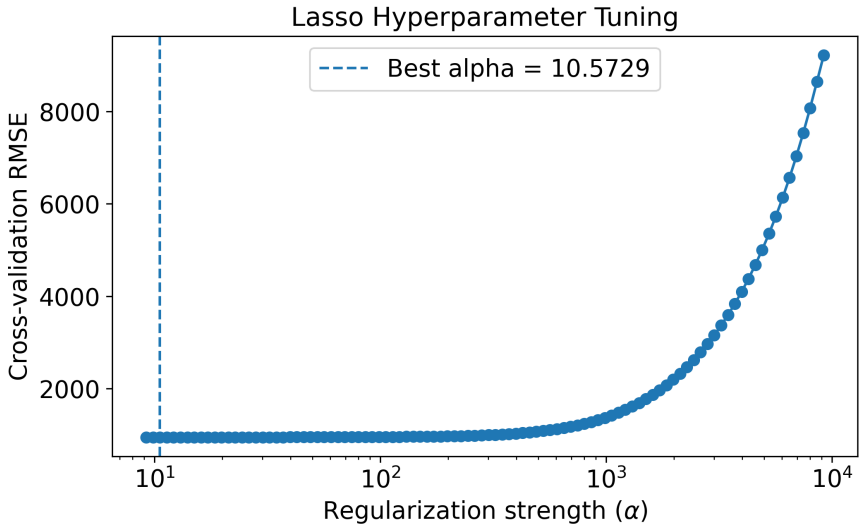


Figure 11: Cross-validation RMSE for Lasso without PCA.

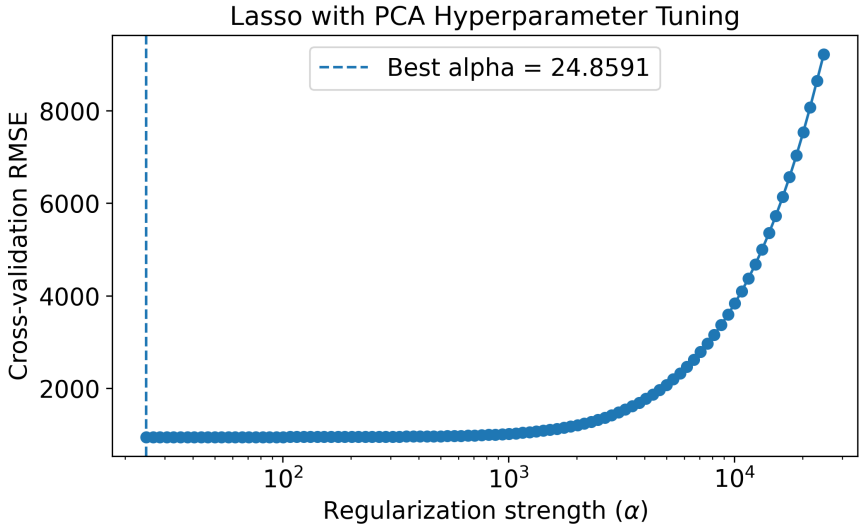


Figure 12: Cross-validation RMSE for Lasso with PCA.

5.4 Support Vector Regression

Support Vector Regression (SVR) with a radial basis function kernel was used to capture nonlinear relationships. The SVR pipeline standardized input features and standardized the target during fitting. Predictions were transformed back to the original target scale, so RMSE and MAE are reported in kilograms.

Hyperparameters were tuned using five-fold cross-validation and Grid Search over 27 combinations of C, ϵ, and γ. The configuration with the lowest cross-validation RMSE was selected. Tuning was performed separately for the feature matrices with and without PCA.

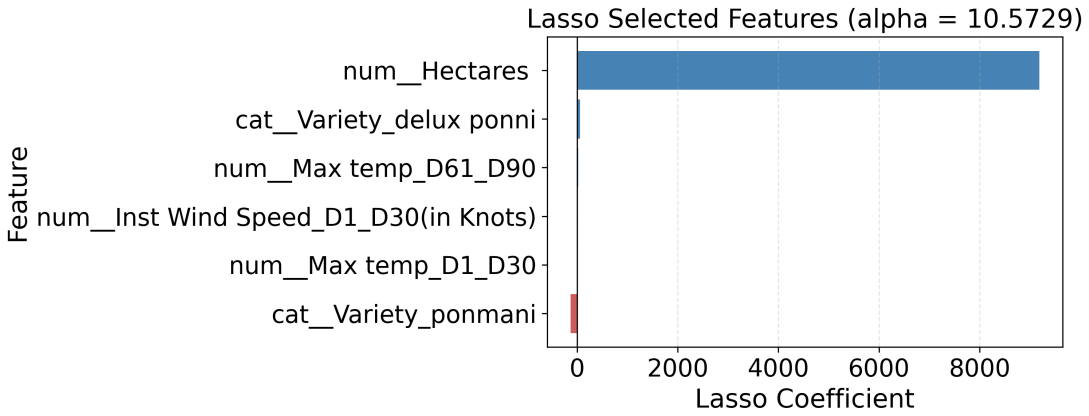


Figure 13: Selected Lasso coefficients without PCA.

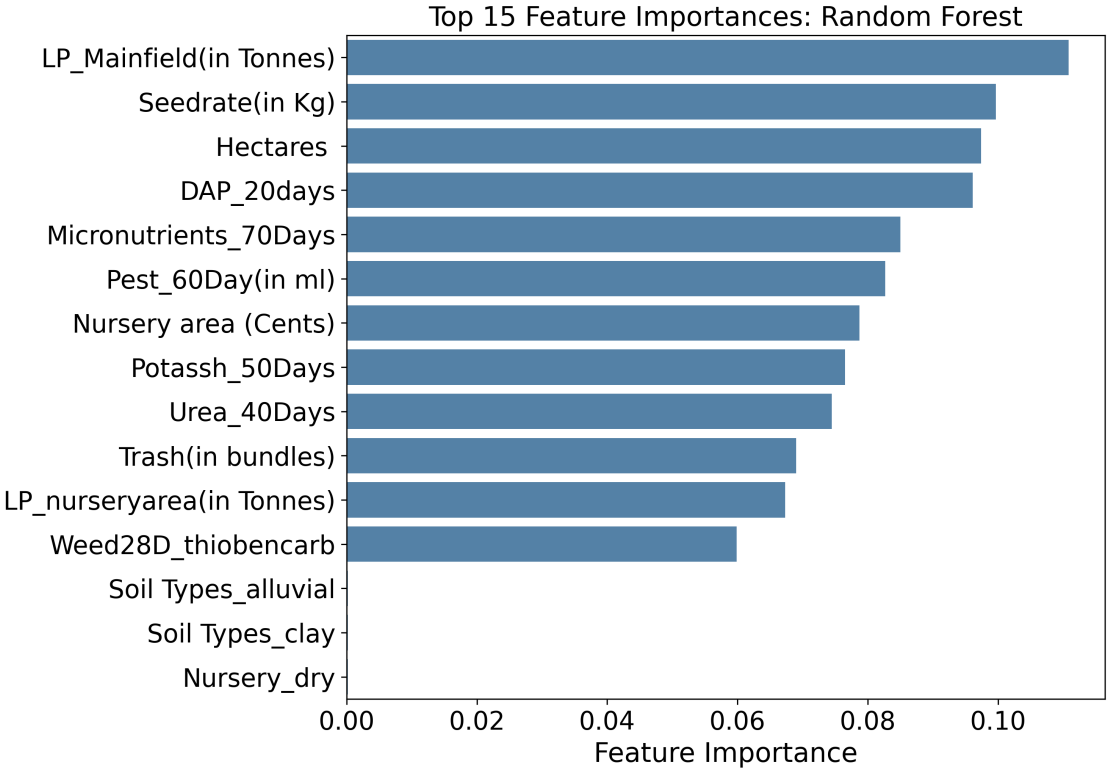


Figure 14: Top feature importances in Random Forest without PCA.

For SVR without PCA, the best hyperparameters were $C = 10$, $\epsilon = 0.01$, and $\gamma = 0.01$, with a cross-validation RMSE of 842.8851 kg. The model achieved a test RMSE of 870.3701 kg, an MAE of 611.7301 kg, and an R^2 of 0.9907, with a training time of 90.2295 seconds.

For SVR with PCA, the best hyperparameters were $C = 100$, $\epsilon = 0.01$, and $\gamma = 0.1$, with a cross-validation RMSE of 873.4547 kg. The model achieved a test RMSE of 882.2195 kg, an MAE of 595.4466 kg, and an R^2 of 0.9904, with a training time of 27.1159 seconds.

5.5 Model Comparison

Table 1 compares the eight model configurations using RMSE, MAE, R^2, and training time. RMSE and MAE are measured in kilograms, while R^2 summarizes the proportion of target variance explained on the test set.

Table 1: Model performance on the held-out test set.

Model	RMSE	MAE	R^2	Time (s)
SVR (without PCA)	**870.3701**	611.7301	**0.9907**	113.3135
SVR (with PCA)	882.2195	**595.4466**	0.9904	27.1159
Random Forest (without PCA)	886.8067	632.5830	0.9903	3.6524
Random Forest (with PCA)	887.6489	633.5083	0.9903	2.0543
Linear Regression (without PCA)	927.9360	698.3192	0.9894	0.0318
Lasso (without PCA)	941.0506	706.5258	0.9891	0.3509
Linear Regression (with PCA)	1004.9139	747.8310	0.9875	**0.0048**
Lasso (with PCA)	1006.9530	750.1580	0.9875	0.1043

We see that SVR without PCA achieved the lowest RMSE and highest R^2. SVR with PCA achieved the lowest MAE and reduced the reported training time from 113.3135 to 27.1159 seconds. Random Forest was also competitive and required much less training time than SVR. PCA did not improve predictive accuracy in this experiment in general, but it reduced training time across all four models.

Interestingly, the simpler models, Linear Regression and Lasso Regression, also performed well. This may be because several features, such as cultivated area and agricultural inputs, have strong relationships with the target and can explain a substantial proportion of its variation. As shown in Figure 6, some features also exhibit approximately linear relationships with paddy yield, which may help explain the strong performance of Linear Regression.

Overall, all models achieved high R^2 values, between 0.9875 and 0.9907, but their MAE and RMSE values remained in the hundreds of kilograms. The absolute errors should be interpreted relative to the target's distribution and scale; high R^2 alone does not mean that individual prediction errors are negligible.

6 Conclusions

This project compared Linear Regression, Lasso Regression, Random Forest Regression, and SVR, each with and without PCA. SVR without PCA achieved the lowest RMSE (870.3701 kg) and highest R^2 (0.9907). SVR with PCA had the lowest MAE (595.4466 kg) and required less training time.

PCA may be useful when predictors are highly correlated or redundant and reducing dimensionality and computational cost is a priority. However, PCA prioritizes variance in the predictors rather than relevance to the target, and its components are less directly interpretable than the original features. Therefore, when understanding the contributions of individual agricultural features is important, using the original features without PCA may be preferable.

The EDA and model importance analyses repeatedly highlighted cultivated area, seeding rate, and agricultural inputs. These findings represent associations rather than causal effects. In particular, the role of `Hectares` depends on the target definition: if the target is total production in kilograms, larger fields can naturally produce more rice.

A Code Repository

The code repository for this project is available in the GitHub repository: https://github.com/ronggan01/Machine-Learning-in-Finance-Fall-2026

References

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